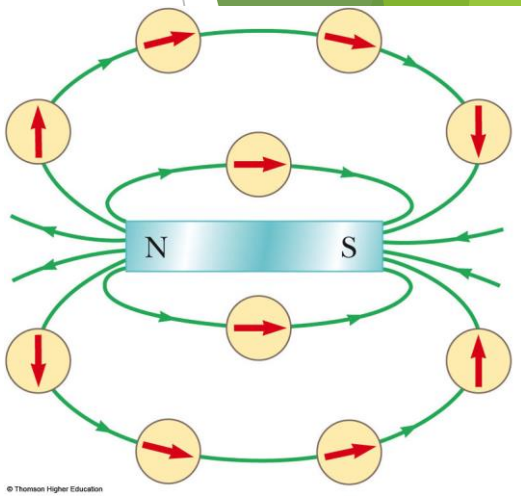
Chapter 29
Magnetic Fields

Sec 29.1: Analysis Model: Particle in a Field (Magnetic)

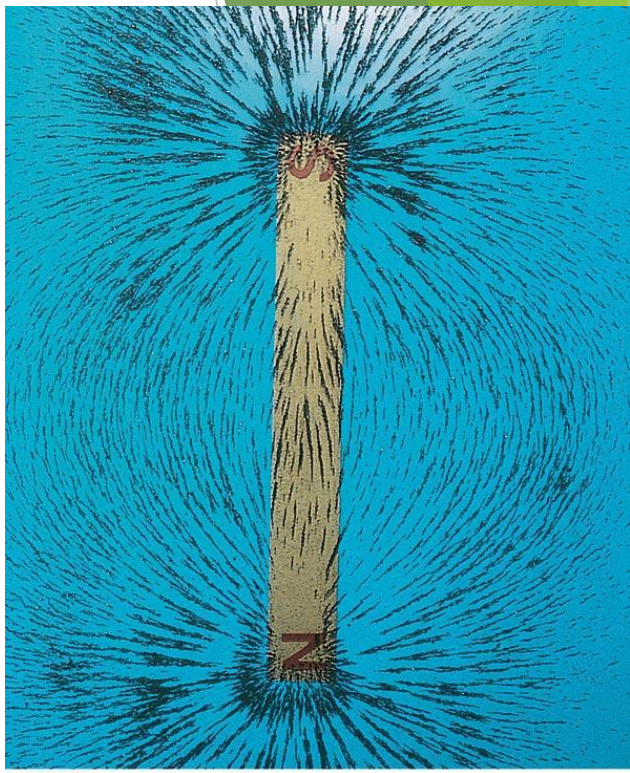
- Every magnet, regardless of its shape, has two poles
  - Called north and south poles
  - Magnetic poles are always found in pairs
  - Poles exert forces on one another
    - Like poles repel each other N-N or S-S
    - Unlike poles attract each other N-S



- A vector quantity
- Symbolized by $\vec{\mathbf{B}}$
- Direction is given by the direction a north pole of a compass needle points in that location
- Magnetic field lines can be used to show how the field lines, as traced out by a compass, would look
- The compass can be used to trace the field lines
- The lines outside the magnet point from the North pole to the South pole

Magnetic Field Lines, Bar Magnet

- Iron filings are used to show the pattern of the magnetic field lines
- The direction of the field is the direction a north pole would point
- The magnetic field at some point in space can be defined in terms of the magnetic force, $\vec{\mathbf{F}}_B$
- The magnetic force will be exerted on a charged particle moving with a velocity, $\vec{\mathbf{v}}$
  - Assume (for now) there are no gravitational or electric fields present



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Force on a Charge Moving in a Magnetic Field

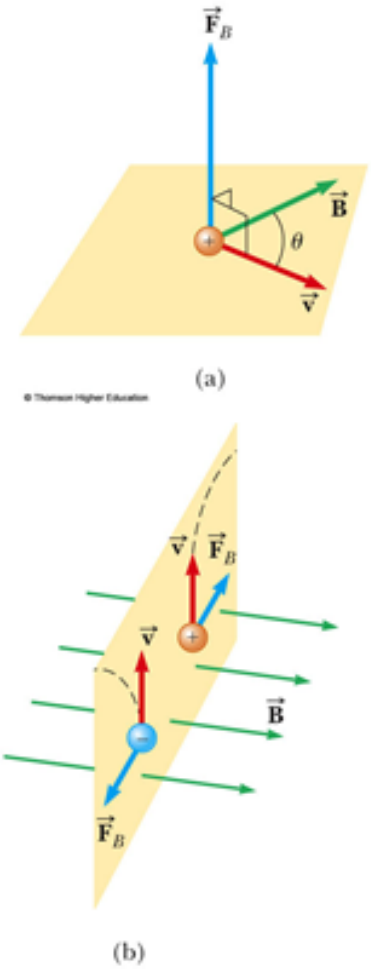
- The magnitude F_B of the magnetic force exerted on the particle is proportional to the charge, q, and to the speed, v, of the particle
- When a charged particle moves parallel to the magnetic field vector, the magnetic force acting on the particle is zero
- When the particle's velocity vector makes any angle $\theta \neq 0$ with the field, the force acts in a direction perpendicular to both the velocity and the field
- The magnetic force exerted on a positive charge is in the direction opposite the direction of the magnetic force exerted on a negative charge moving in the same direction
- The magnitude of the magnetic force is proportional to sin θ, where θ is the angle the particle's velocity makes with the direction of the magnetic field

Force on a Charge Moving in a Magnetic Field

- $\vec{\mathbf{F}}_B$ is perpendicular to the plane formed by $\vec{\mathbf{v}}$ and $\vec{\mathbf{B}}$
- Oppositely directed forces exerted on oppositely charged particles will cause the particles to move in opposite directions
- The properties can be summarized in a vector equation:

$$\vec{\mathbf{F}}_B = q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$

  - $\vec{\mathbf{F}}_B$ is the magnetic force
  - q is the charge
  - $\mathbf{v}$ is the velocity of the moving charge
  - $\vec{\mathbf{B}}$ is the magnetic field

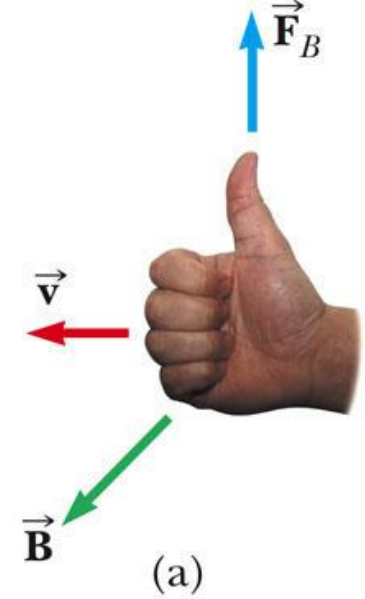


More About Magnitude of F

- The magnitude of the magnetic force on a charged particle is $F_B = |q|\ v\ B \sin\theta$
  - θ is the smaller angle between v and B
  - F_B is zero when the field and velocity are parallel or antiparallel
    - $\theta = 0$ or 180°
  - F_B is a maximum when the field and velocity are perpendiculars
    - $\theta = 90^o$

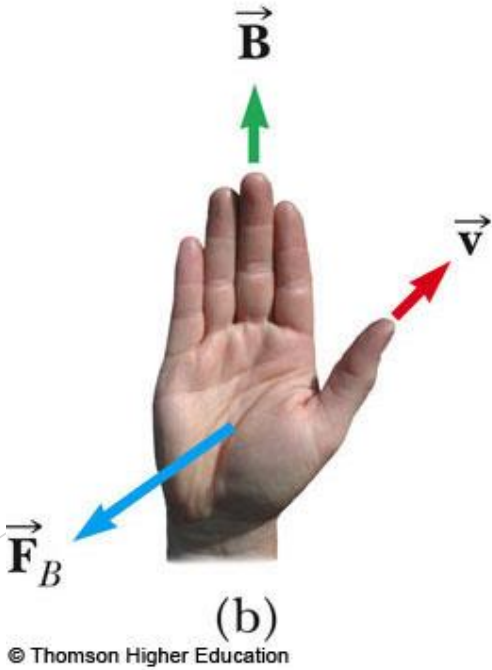
Direction: Right-Hand Rule #1

- The fingers point in the direction of $\vec{\mathbf{v}}$
- $\vec{\mathbf{B}}$ comes out of your palm
  - Curl your fingers in the direction of $\vec{\mathbf{B}}$
- The thumb points in the direction of $\vec{\mathbf{v}} \times \vec{\mathbf{B}}$ which is the direction of $\vec{\mathbf{F}}_B$



Direction: Right-Hand Rule #2

- Alternative to Rule #1
- Thumb is in the direction of $\vec{\mathbf{v}}$
- Fingers are in the direction of $\vec{\mathbf{B}}$
- Palm is in the direction of $\vec{\mathbf{F}}_B$
  - On a positive particle
  - You can think of this as your hand pushing the particle



Differences Between Electric and Magnetic Fields

- Direction of force
  - The electric force acts along the direction of the electric field
  - The magnetic force acts perpendicular to the magnetic field
- Motion
  - The electric force acts on a charged particle regardless of whether the particle is moving
  - The magnetic force acts on a charged particle only when the particle is in motion
- Work
  - The electric force does work in displacing a charged particle
  - The magnetic force associated with a steady magnetic field does no work when a particle is displaced
    - This is because the force is perpendicular to the displacement

Work in Fields

- The kinetic energy of a charged particle moving through a magnetic field cannot be altered by the magnetic field alone
- When a charged particle moves with a given velocity through a magnetic field, the field can alter the direction of the velocity, but not the speed or the kinetic energy
- The SI unit of magnetic field is the tesla (T)

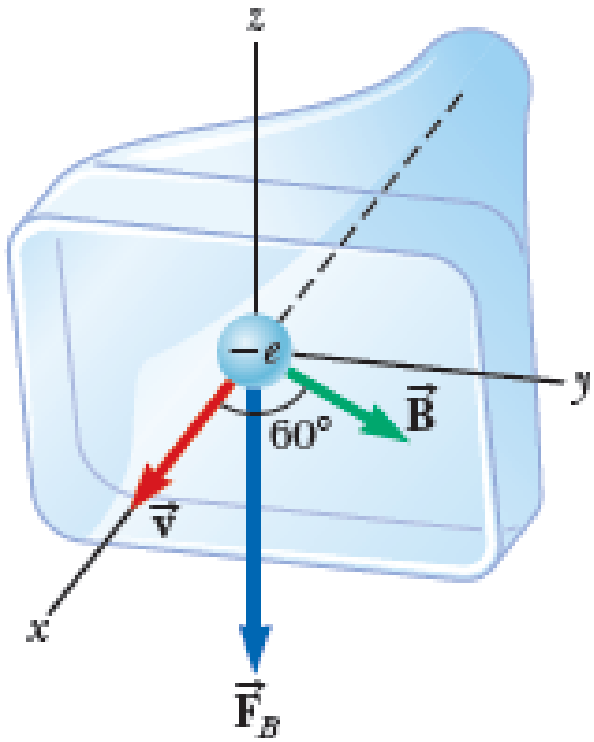
$$T = \frac{Wb}{m^2} = \frac{N}{C \cdot (m/s)} = \frac{N}{A \cdot m}$$

- Wb is a weber
- A non-SI commonly used unit is a gauss (G)
- 1 T = 104 G

Example 29.1 An Electron Moving in a Magnetic Field

AM

An electron in an old-style television picture tube moves toward the front of the tube with a speed of 8.0×10^6 m/s along the x axis (Fig. 29.6). Surrounding the neck of the tube are coils of wire that create a magnetic field of magnitude 0.025 T, directed at an angle of 60° to the x axis and lying in the xy plane. Calculate the magnetic force on the electron.



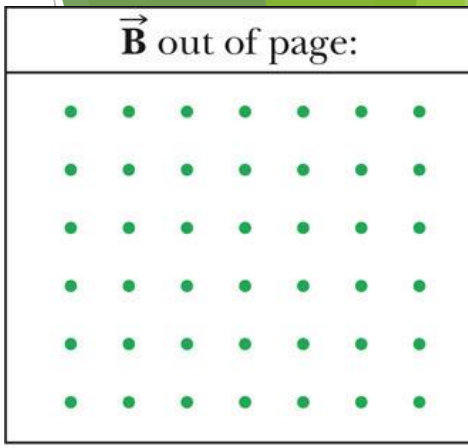
$$F_B = |q|vB\sin\theta$$

$$= (1.6 \times 10^{-19}\ \text{C})(8.0 \times 10^6\ \text{m/s})(0.025\ \text{T})(\sin 60°)$$

$$= \boxed{2.8 \times 10^{-14}\ \text{N}}$$

Sec 29.2: Motion of a Charged Particle in a Uniform Magnetic Field

- When vectors are perpendicular to the page, dots, and crosses are used
  - The dots represent the arrows coming out of the page
  - The crosses represent the arrows going into the page
- Consider a particle moving in an external magnetic field with its velocity perpendicular to the field
- The force is always directed toward the center of the circular path
- The magnetic force causes a centripetal acceleration, changing the direction of the velocity of the particle



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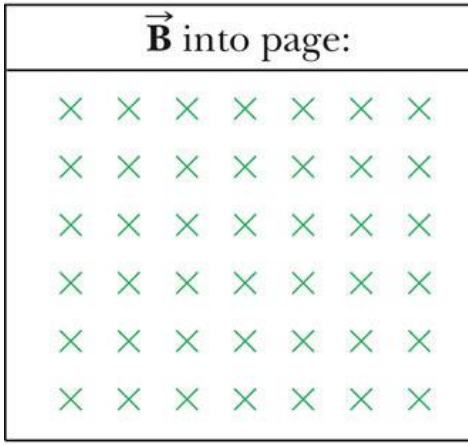

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Charged Particle in a Magnetic Field

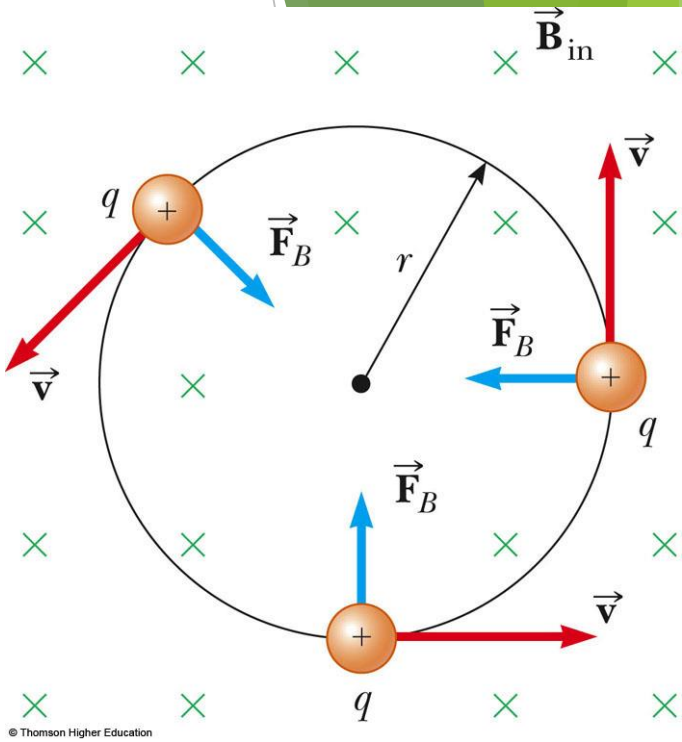
- Equating the magnetic and centripetal forces:

$$F_B = qvB = \frac{mv^2}{r}$$

- Solving for r: $$r = \frac{mv}{qB}$$
  - r is proportional to the linear momentum of the particle and inversely proportional to the magnetic field
- The angular speed of the particle is

$$\omega = \frac{v}{r} = \frac{qB}{m}$$

  - The angular speed, ω, is also referred to as the cyclotron frequency
- The period of the motion is $$T = \frac{2\pi r}{v} = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}$$



Example 29.2 A Proton Moving Perpendicular to a Uniform Magnetic Field

AM

A proton is moving in a circular orbit of radius 14 cm in a uniform 0.35-T magnetic field perpendicular to the velocity of the proton. Find the speed of the proton.

$$v = \frac{qBr}{m_p}$$

$$v = \frac{(1.60 \times 10^{-19}\ \mathrm{C})(0.35\ \mathrm{T})(0.14\ \mathrm{m})}{1.67 \times 10^{-27}\ \mathrm{kg}}$$

$$= \boxed{4.7 \times 10^{6}\ \mathrm{m/s}}$$

Example 29.3 Bending an Electron Beam AM

In an experiment designed to measure the magnitude of a uniform magnetic field, electrons are accelerated from rest through a potential difference of 350 V and then enter a uniform magnetic field that is perpendicular to the velocity vector of the electrons. The electrons travel along a curved path because of the magnetic force exerted on them, and the radius of the path is measured to be 7.5 cm. (Such a curved beam of electrons is shown in Fig. 29.10.)

(A) What is the magnitude of the magnetic field?

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$$\Delta K + \Delta U = 0$$

$$\left(\tfrac{1}{2}m_e v^2 - 0\right) + (q\,\Delta V) = 0$$

$$v = \sqrt{\frac{-2q\,\Delta V}{m_e}}$$

$$v = \sqrt{\frac{-2(-1.60 \times 10^{-19}\ \mathrm{C})(350\ \mathrm{V})}{9.11 \times 10^{-31}\ \mathrm{kg}}} = 1.11 \times 10^{7}\ \mathrm{m/s}$$

$$B = \frac{m_e v}{er}$$

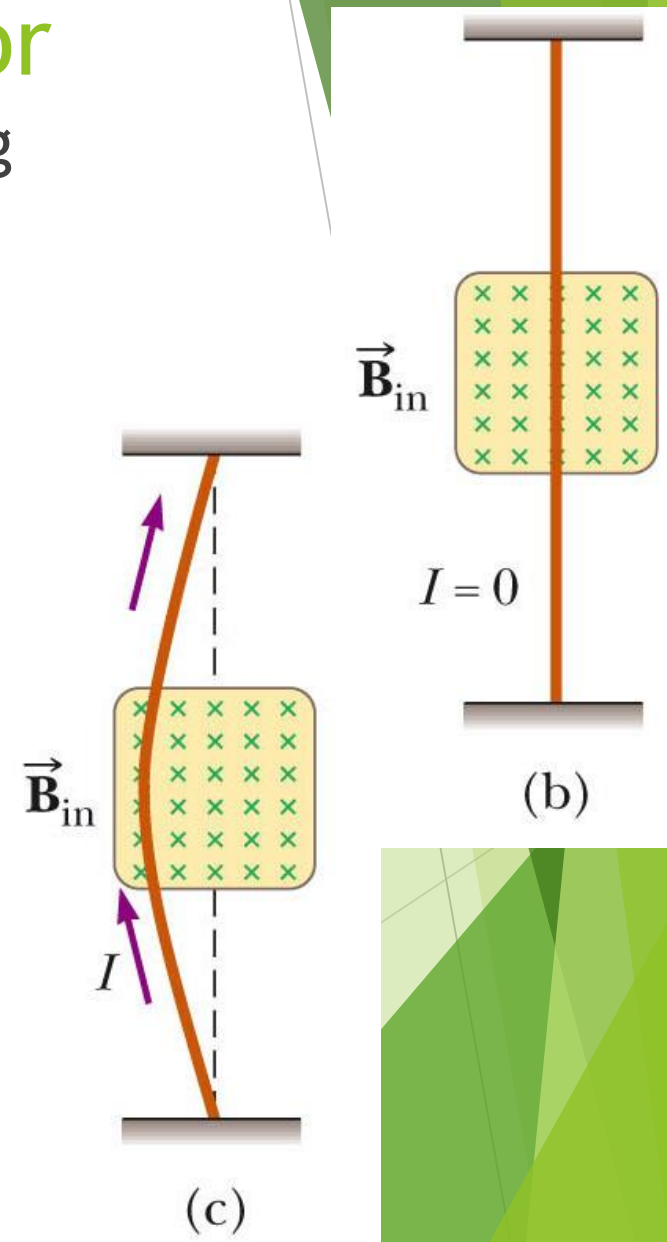
$$B = \frac{(9.11 \times 10^{-31}\ \mathrm{kg})(1.11 \times 10^{7}\ \mathrm{m/s})}{(1.60 \times 10^{-19}\ \mathrm{C})(0.075\ \mathrm{m})} = 8.4 \times 10^{-4}\ \mathrm{T}$$

(B) What is the angular speed of the electrons?

$$\omega = \frac{v}{r} = \frac{1.11 \times 10^{7}\ \mathrm{m/s}}{0.075\ \mathrm{m}} = 1.5 \times 10^{8}\ \mathrm{rad/s}$$

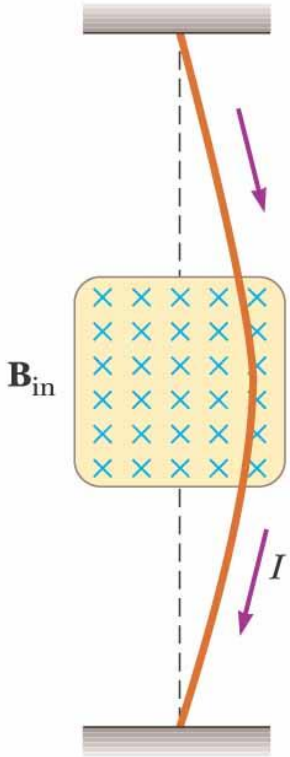
Sec 29.4: Magnetic Force Acting on a Current- Carrying Conductor

- A force is exerted on a current-carrying wire placed in a magnetic field
  - The current is a collection of many charged particles in motion
- The direction of the force is given by the right-hand rule
- In this case, there is no current, so there is no force, Therefore, the wire remains vertical
- The magnetic field is into the page
- The current is up the page
- The force is to the left



Force on a Wire

- The magnetic field is into the page
- The current is down the page
- The force is to the right



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- In terms of the current:

$$\vec{\mathbf{F}}_B = I\vec{\mathbf{L}} \times \vec{\mathbf{B}}$$

  - I is the current
  - $\vec{\mathbf{L}}$ is a vector that points in the direction of the current
    - Its magnitude is the length L of the segment
  - $\vec{\mathbf{B}}$ is the magnetic field